

SENTIMEN ANALISIS BERBASIS ASPEK PADA PERUMDA AIR MINUM SURYA SEMBADA SURABAYA MENGGUNAKAN KOMPUTASI KORELASI

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OBJECTIVES

1

The system is expected to classify comments based on 10 topics without manual intervention.

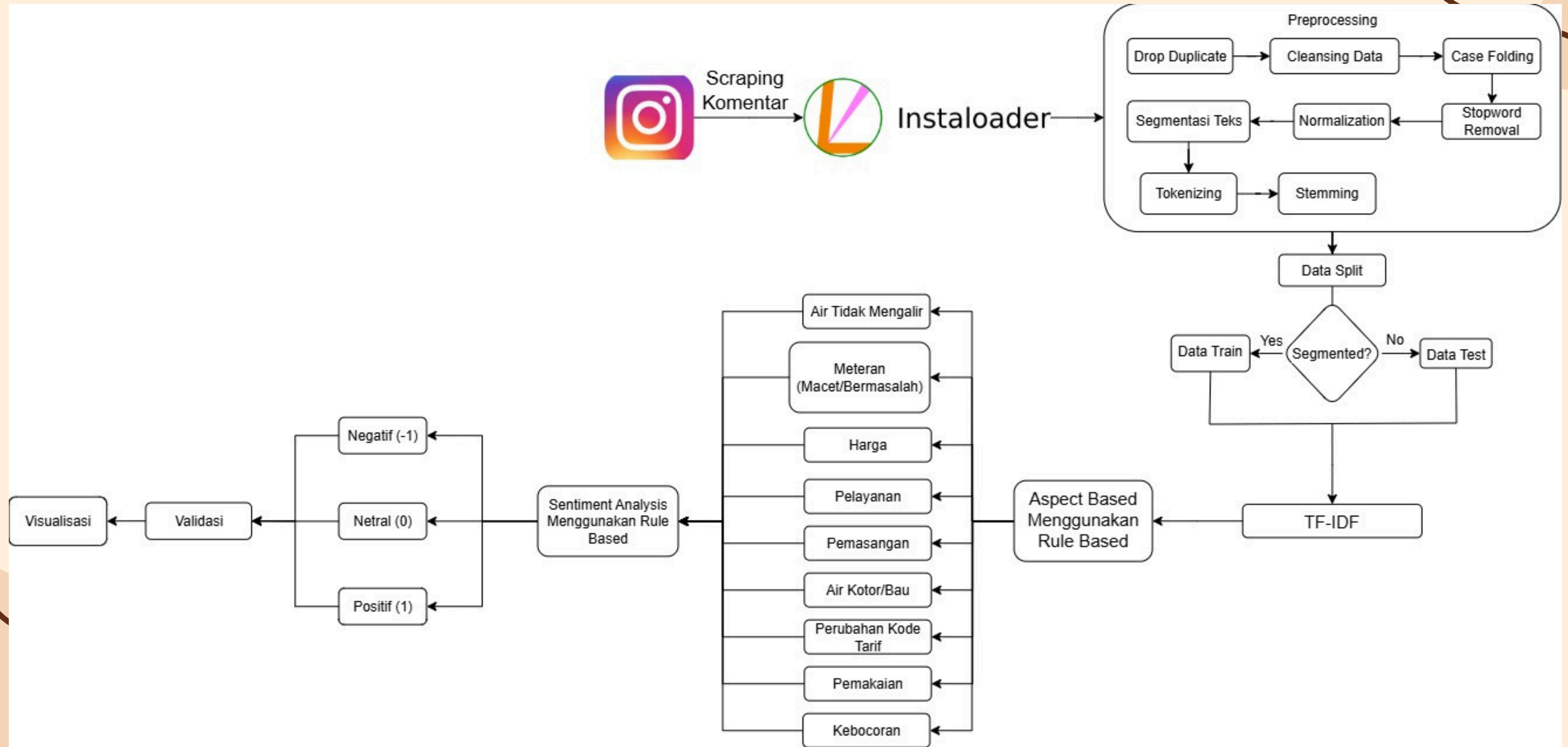
2

It should be able to understand whether comments express positive, negative, or neutral feelings toward PDAM services.

3

By using cosine similarity and rule-based sentiment analysis, the system is expected to perform better for short, unstructured comment data.

SYSTEM DESIGN



PREPROCESSING DATA

Process	Input		Output	
	Sentence 1	Sentence 2	Sentence 1	Sentence 2
Case Folding	Pemasangan baru tidak ada kelanjutan setelah survey. Padahal katanya bisa komplain CS lewat WA. Sedangkan di WA gak ada tanggapan		pemasangan baru tidak ada kelanjutan setelah survey. padahal katanya bisa komplain cs lewat wa. sedangkan di wa gak ada tanggapan	
	Min kalau puteran PDAM dol itu sya ngadu kesiapa ya		min kalau puteran pdam dol itu sya ngadu kesiapa ya	

PREPROCESSING DATA

Process	Input		Output	
	Sentence 1	Sentence 2	Sentence 1	Sentence 2
Normalization	pemasangan baru tidak ada kelanjutan setelah survey. padahal katanya bisa komplain cs lewat wa. sedangkan di wa gak ada tanggapan		pemasangan baru tidak ada kelanjutan setelah survey. padahal katanya bisa komplain cs lewat wa. sedangkan di wa gak ada tanggapan	
	min kalau puteran pdam dol itu sya ngadu kesiapa ya		admin kalau keran pdam rusak itu saya mengadu ke siapa ya	

PREPROCESSING DATA

Process	Input		Output	
	Sentence 1	Sentence 2	Sentence 1	Sentence 2
Stop Word Removal	pemasangan baru tidak ada kelanjutan survey. padahal katanya bisa komplain pelayanan pelanggan lewat whatsapp. sedangkan di whatsapp tidak ada tanggapan.		pemasangan tidak ada kelanjutan survey. padahal katanya komplain pelayanan pelanggan whatsapp. sedangkan whatsapp tidak ada tanggapan.	
	admin kalau keran pdam rusak itu saya mengadu ke siapa ya		admin keran pdam rusak mengadu ke siapa	

PREPROCESSING DATA

Process	Input		Output	
	Sentence 1	Sentence 2	Sentence 1	Sentence 2
Text Segmentation	pemasangan tidak ada kelanjutan survey. padahal katanya komplain pelayanan pelanggan whatsapp. sedangkan whatsapp tidak ada tanggapan.		pemasangan tidak ada kelanjutan survey.	padahal katanya komplain pelayanan pelanggan whatsapp. sedangkan whatsapp tidak ada tanggapan.
	admin keran pdam rusak mengadu ke siapa		admin keran pdam rusak mengadu ke siapa	

PREPROCESSING DATA

Process	Input		Output	
	Sentence 1	Sentence 2	Sentence 1	Sentence 2
Tokenizing	pemasangan tidak ada kelanjutan survey.	padahal katanya komplain pelayanan pelanggan whatsapp. sedangkan whatsapp tidak ada tanggapan.	['pemasangan', 'tidak', 'ada', 'kelanjutan', 'survey', '.'],	['padahal', 'katanya', 'komplain', 'pelayanan', 'pelanggan', 'whatsapp', '.', 'sedangkan', 'whatsapp', 'tidak', 'ada', 'tanggapan', '.']
	admin keran pdam rusak mengadu ke siapa	[]	['admin', 'keran', 'pdam', 'rusak', 'mengadu', 'ke', 'siapa']	[]

PREPROCESSING DATA

Process	Input		Output	
	Sentence 1	Sentence 2	Sentence 1	Sentence 2
Stemming	['pemasangan', 'tidak', 'ada', 'kelanjutan', 'survey', '.'],	['padahal', 'katanya', 'komplain', 'pelayanan', 'pelanggan', 'whatsapp', '.', 'sedangkan', 'whatsapp', 'tidak', 'ada', 'tanggapan', '.']	['pasang', 'tidak', 'ada', 'lanjut', 'survey', '.']	['padahal', 'kata', 'komplain', 'layan', 'langgan', 'whatsapp', '.', 'sedang', 'whatsapp', 'tidak', 'ada', 'tanggap', '.']
	admin keran pdam rusak mengadu ke siapa	[]	['admin', 'keran', 'pdam', 'rusak', 'adu', 'ke', 'siapa']	[]

Conjunction
Sedang
Sedangkan
Lalu
Tapi
Karena
Tetapi
Padahal
Akan
Sehingga

TEXT SEGMENTATION

The text segmentation process works as expected by splitting comments into two parts based on predefined conjunctions. The system first loads a list of conjunctions, then identifies the earliest one in each normalized comment. If a conjunction is found, the text is divided into sentence_1 (before the conjunction) and sentence_2 (from the conjunction to the end). If no conjunction is present, the entire comment remains in sentence_1 while sentence_2 is left empty. These conjunctions were selected because they commonly connect different ideas in customer comments, helping the system detect context shifts. Evaluation on 1,670 comments showed that 293 contained conjunctions, and all were segmented correctly, resulting in 100% accuracy. This indicates that the conjunction-based segmentation method performs very effectively for the dataset used.

DATA SPLIT

In the data splitting stage, the division was performed after preprocessing. All 1,670 available comments were used as test data to evaluate the system on both single-context and multi-context customer feedback. Meanwhile, the training data consisted only of comments that were not segmented, meaning they contained a single sentence (where `sentence_2` is empty), resulting in 1,377 training instances. This approach ensures that each training sample represents a clear and stable single context, making it more suitable as a reference for aspect classification. The segmented test data are then compared against this unsegmented training set to identify the most relevant aspect for each segment. Overall, this setup supports a more focused multi-aspect classification process by training on simple, single-context data and testing on the full range of comments.

TF-IDF RESULT

	Input	Vektor
<i>Sentence 1</i>	['pasang', 'tidak', 'ada', 'lanjut', 'survey']	0.304617 0.215181 0.240960 0.431406 0.441414
<i>Sentence 2</i>	['padahal', 'kata', 'komplain', 'layan', 'langgan', 'whatsapp', 'sedang', 'whatsapp', 'tidak', 'ada', 'tanggap']	- 0.317079 0.198214 0.351224 0.241194 0.557980 - 0.557980 0.132566 0.148448 0.275340

TF-IDF

The TF-IDF testing phase is conducted to verify whether the system can successfully transform input text into numerical vector representations. Based on the results shown in the previous slide, the vectorization process using the Term Frequency Inverse Document Frequency (TF-IDF) method has performed effectively. This is demonstrated by the system's ability to convert the given text inputs into meaningful decimal vectors.

Aspect Name
Air Tidak Mengalir
Meteran (Macet/Bermasalah)
Harga
Pelayanan
Pemasangan

Aspect Name
Air Kotor/Bau
Kebocoran
Pemakaian
Perubahan Kode Tarif
Lainnya

ASPECT USED

In the aspect-based stage, the author defined 10 aspects as shown in Table:

- **These aspects were determined based on two main sources: the aspects already available in the PDAM CIS application (Air Kotor/Bau, Kebocoran, Meteran (Macet/Bermasalah), Air Tidak Mengalir, Pemakaian, Perubahan Kode Tarif)**
- **The author's own observations after reviewing the dataset (Pemasangan, Pelayanan, Harga, Lainnya).**

ASPECT BASED USING COSINE SIMILARITY

The aspect classification process uses a multi-aspect approach based on cosine similarity. After preprocessing steps- normalization, text segmentation, tokenizing, and stemming- each comment is converted into a TF-IDF vector. The training data consists of single-segment comments (without sentence_2), while all data is used for testing. Each test segment (sentence_1 and sentence_2) is compared with all training data to find the aspect with the highest similarity score. If sentence_2 is empty, only sentence_1 is used for prediction. Since segmentation is limited to two parts, the system can assign a maximum of two aspects per comment. For single-aspect classification, cosine similarity is calculated between the first segment vector and all training vectors using the defined formula.

COSINE SIMILARITY FORMULA

$$\text{Sim}(S1, Di) = \frac{(S1 \cdot Di)}{(|S1| \times |Di|)}$$

Description:

- **Sim(S1, Di):** The cosine similarity value between sentence_1 and the i-th training data.
- **S1:** The TF-IDF vector representation of sentence_1.
- **Di:** The TF-IDF vector representation of the i-th training data.
- **S1 · Di:** The dot product of the two vectors, calculated by multiplying the TF-IDF weights of matching features.
- **|S1|:** The magnitude (length) of the sentence_1 vector, computed as the square root of the sum of the squares of all its elements.
- **|Di|:** The magnitude (length) of the i-th training vector, computed as the square root of the sum of the squares of all its elements.

EXAMPLE OF ASPECT CALCULATION

The aspect with the highest similarity score from the training data is selected as the primary candidate. If the top similarity score for sentence_1 is at least 0.30 and not classified as "Lainnya," it is assigned as the first aspect; if it is below 0.18, the comment is labeled as "Lainnya." The first aspect comes from sentence_1, while the second aspect is taken from sentence_2 if its similarity score is at least 0.40, not "Lainnya," and different from the first aspect. If no aspects meet these criteria, the system selects the highest similarity score from both segments, and if this score is still below 0.18, the final result is "Lainnya."

EXAMPLE OF ASPECT CALCULATION

Word Table

	Words
Sentence 1	['pasang', 'tidak', 'ada', 'lanjut', 'survey']
Data Train	['pasang', 'tidak', 'ada', 'biaya', 'tambah']

TF-IDF Calculation Table

	pasang	tidak	ada	pasang tidak	tidak ada
<i>Sentence 1</i>	0.304617	0.215181	0.240960	0.547104	0.350091
<i>Data Train</i>	0.267715	0.189113	0.211769	0.480826	0.307680

EXAMPLE OF ASPECT CALCULATION

The word table represents the sentence used in the similarity calculation for sentence_1, while the TF-IDF calculation table shows the TF-IDF results for both sentence_1 and the training data. The terms listed from the training data are those with the highest similarity to sentence_1 under the Pemasangan aspect. The presence of two word features such as “pasang tidak” and “tidak ada” occurs because the vectorizer uses an ngram_range of (1,2), meaning it captures both single words (unigrams) and consecutive word pairs (bigrams). This allows the model to better capture local context during classification.

EXAMPLE OF ASPECT CALCULATION

$$\text{Sim}(S1, D1180) = \frac{((0.304617 \times 0.267715) + (0.215181 \times 0.189113) + (0.240960 \times 0.211769) + (0.547104 \times 0.480826) + (0.350091 \times 0.307680))}{\sqrt{0.304617^2 \times 0.215181^2 \times 0.240960^2 \times 0.547104^2 \times 0.350091^2} \times \sqrt{0.267715^2 \times 0.189113^2 \times 0.211769^2 \times 0.480826^2 \times 0.307680^2}}$$

$$\text{Sim}(S1, D1180) = 0.544049$$

The similarity score of 0.544049 exceeds the main threshold of 0.30, so sentence_1 is classified under the Pemasangan aspect.

ASPECT BASED MODEL EVALUATION

The experimental results show that the system successfully classified aspects for 1,670 test data using 1,377 training data, effectively handling both single-aspect and multi-aspect comments. The most frequently predicted aspects were *Lainnya*, *Air Tidak Mengalir*, *Meteran (Macet/Bermasalah)*, and *Harga*, indicating that the system can map customer comments to appropriate service aspects, even when multiple contexts are present. Overall, the system achieved an accuracy of 89.28%, with 1,491 correct predictions and 179 misclassifications, along with an F1-score of 0.9321, demonstrating strong performance. However, a limitation remains as the system can only process up to two aspects per comment, based on the segmentation approach used.

SENTIMENT ANALYSIS USING RULE BASED

In this stage, sentiment analysis is performed using a rule-based approach that combines part-of-speech (POS) patterns and a predefined polarity dictionary. Unlike aspect classification, which uses stemmed text, this process uses `normalized_text` to preserve important domain-specific meanings. The workflow begins by cleaning and splitting the normalized text into sentence fragments, followed by phrase detection, POS matching, and sentiment scoring based on defined rules. The scores from each fragment are summed to determine the final sentiment: positive (greater than 0), neutral (equal to 0), or negative (less than 0). Additional adjustments include assigning neutral sentiment to questions without strong negative signals, recognizing strong negative phrases (e.g., *tidak ada respon*, *air tidak mengalir*), and applying a complaint context penalty when complaint-related and service-related terms appear together. The system also uses AND and NAND operators to evaluate word pairs, helping refine sentiment calculations based on their combined values.

RULE EXPLANATION

1. Single Adjective

The first rule is single adjectives. If a sentence contains an adjective with a sentiment value and it is not involved in any other rule combination, its sentiment value is directly used as the sentence score.

Example: the word lama is an adjective with a value of -1 →
score = sentiment(lama) = -1 → Negative

2. Single Verb

The second rule is single verbs. If a sentence contains a verb with a sentiment value and it is not paired with other rules, its value is directly applied.

Example: the word bayar is a verb with a value of -1 →
score = sentiment(bayar) = -1 → Negative

RULE EXPLANATION

3. Preposition + Adjective

This rule combines a preposition and an adjective using the AND operator.

Example: air dalam keruh $\rightarrow$ dalam = 1, keruh = -1 $\rightarrow$ AND(1, -1) = -1

The word air also has a value of -1 $\rightarrow$

score = sentiment(air) + AND(dalam, keruh) = -1 + (-1) = -2 $\rightarrow$ Negative

4. Preposition + Verb

This rule combines a preposition and a verb, often appearing in negation forms.

Example: tidak ada kelanjutan $\rightarrow$ complaint penalty = -1, tidak = -1, ada = 0 $\rightarrow$

score = complaint penalty + sentiment(tidak) + sentiment(ada) = -1 + (-1) + 0 = -2 $\rightarrow$

Negative

RULE EXPLANATION

5. Verb + Adjective

This rule combines a verb and an adjective using the NAND operator.

Example: bayar mahal $\rightarrow$ bayar = -1, mahal = -1 $\rightarrow$ NAND(-1, -1) = -1

The clause also includes air keruh = -1 and bau = -1 $\rightarrow$

score = NAND(bayar, mahal) + sentiment(air keruh) + sentiment(bau) = -1 + (-1) + (-1)
= -3 $\rightarrow$ Negative

6. Preposition + Verb + Adjective

This rule combines a preposition, verb, and adjective.

Example: air tidak keluar mati $\rightarrow$ tidak keluar = -1, mati = -1 $\rightarrow$

score = sentiment(tidak keluar) + sentiment(mati) = -1 + (-1) = -2 $\rightarrow$ Negative

RULE EXPLANATION

7.Fixed Phrase (Two Words, Single Meaning)

The system also supports phrase-based rules, where certain phrases are treated as a single unit of meaning.

Example: terima kasih →

score = sentiment(terima kasih) = 1 → Positive

8.Strong Negative Phrase

The model recognizes specific phrases as strongly negative.

Example: tidak ada tanggapan →

score = sentiment(tidak ada tanggapan) = -1 → Negative

RULE EXPLANATION

9.Question Rule

The system assigns a neutral sentiment to questions without strong negative signals.

Example: pdam naik selalu? →

score = 0 → Neutral

10.Complaint Context Penalty

The system applies a penalty when detecting complaint-related context.

Example: aliran air tidak lancar → complaint penalty = -1, air = -1, tidak lancar = -1 →

score = $-1 + (-1) + (-1) = -3$ → Negative

SENTIMENT ANALYSIS EVALUATION

The rule-based sentiment analysis model was tested on 1,670 data points, with original labels consisting of 827 Negative, 568 Neutral, and 275 Positive. The system correctly predicted 1,340 data points and misclassified 330, achieving an accuracy of 80.24%, an error rate of 19.76%, and an F1-score of 0.8027, indicating fairly good performance in identifying sentiment polarity. The confusion matrix shows that most Negative (668), Neutral (447), and Positive (225) data were correctly classified, although errors mainly occurred between Negative and Neutral labels due to ambiguous or indirect complaint expressions. Overall, the model performs reasonably well, with the advantage of being transparent and easy to interpret, but its performance still depends heavily on the completeness of the dictionary, phrase coverage, and rule design.

DONE AND NOT DONE

Done	Not Done
Collecting the dataset	Application development
Text preprocessing	
Data split	
Developing and evaluating text segmentation	
Developing and evaluatiing aspect based using cosine similarity	
Developing and evaluating sentiment analysis using rule based	

CHALLENGES

The current system only splits comments into two segments, so comments containing more than two aspects cannot be processed optimally.

The rule-based approach heavily depends on the completeness of the dictionary, phrases, and defined rules, which makes it less effective for handling informal language, implicit meanings, or ambiguous contexts.

Comments that contain irony, sarcasm, or complex sentence structures also remain challenging, as their sentiment cannot always be accurately captured through direct rule matching.

CONCLUSION

The experimental results show that the proposed solution effectively addresses the main problem of classifying PDAM customer comments into relevant service aspects and automatically identifying their sentiment. The multi-aspect classification stage achieved an accuracy of 89.28% (1,491 correct predictions out of 1,670), demonstrating that the combination of cosine similarity and text segmentation works well for both single- and dual-aspect comments, despite being limited to a maximum of two aspects per comment. Meanwhile, the rule-based sentiment analysis achieved an accuracy of 80.24% and an F1-score of 0.8027, indicating that the use of POS patterns, phrases, negation, and complaint context can reliably capture sentiment polarity. Overall, the system proves to be effective in helping PDAM better understand customer complaints in a more structured way, both in terms of service aspects and sentiment.

THANK
YOU

